SYSTEM BLOCK DIAGRAM

ChryslerScanner V240 | integrated microSD | 329 components incl. 52 test points | 162 connected nets

VEHICLE CONNECTION

BUS INTERFACES

CONTROL & SENSING

HOST & LOGGING

OBD connector + protection

File: 06_obd-protection.kicad_sch

Battery / GND

Vehicle power input

File: 07_vehicle-input.kicad_sch

Protected supply

CCD interface

Bus data

File: 01_ccd.kicad_sch

PCI / J1850

File: 02_pci.kicad_sch

SCI stages

SCI TX14 / TX9

File: 03_sci-14-9.kicad_sch

SCI TX6 / TX12 + RX bias

File: 04_sci-6-12.kicad_sch

SCI RX7 / RX15

File: 05_sci-rx.kicad_sch

Debug access / test points

48 hook loops + 4 SPI pads; named global nets

File: 24_debug-testpoints.kicad_sch

ESP32 + RF

UART / reset

SPI / detect

File: 16_mcu.kicad_sch

I2C

I/O expander + I2C

File: 18_control.kicad_sch

AW9523B | bus enables and programming control

ADC

Voltage monitoring

File: 17_adc.kicad_sch

Enables / bias

USB + reset / boot

File: 19_usb.kicad_sch

J2 USB-C | CP2102 bridge | reset / boot

Integrated SD logger

File: 22_sd-logger.kicad_sch

J4 microSD | SD_3V3 | LOG / STOP

Status LED drivers

File: 20_leds.kicad_sch

Status

Hardware safety + ADC isolation

File: 23_hardware-safety.kicad_sch

POWER GENERATION & SCI PROGRAMMING

5 V / 3.3 V supplies

File: 08_power-5v.kicad_sch

VPP boost supply

File: 09_power-vpp.kicad_sch

VBB supply

File: 10_power-vbb.kicad_sch

7.5 V supply

File: 11_power-7v5.kicad_sch

SCI TX6 VBB / VPP drivers

File: 12_program-6.kicad_sch

SCI TX9 VBB / VPP drivers

File: 13_program-9.kicad_sch

SCI TX12 VBB / VPP drivers

File: 14_program-12.kicad_sch

SCI TX14 VBB / VPP drivers

File: 15_program-14.kicad_sch

Separate VBB / VPP rails feed the SCI programming stages; enables come from AW9523B.

Solid arrows: data and user interface paths. Dashed arrows: power or control relationships.

Double-click a block to open its circuit. Native hierarchical pins and named stubs carry the exact design nets.

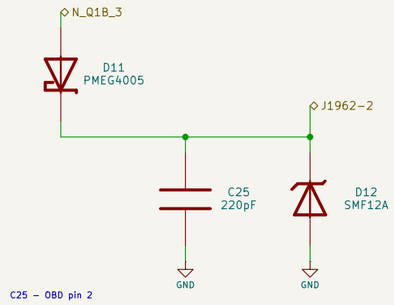
J1 and input protection share one sheet. Solid arrows show data flow; native named stubs define electrical connectivity.

ChryslerScanner / atopile	
Sheet: /	
File: ChryslerScanner_V240_SD.kicad_sch	
Title: ChryslerScanner V240 — system block diagram	
Size: A1 | Date: 2026-10-21	Rev: V240-SD-C
KiCad E.D.A. 10.0.5	Id: 1/24

OBD connector and line protection

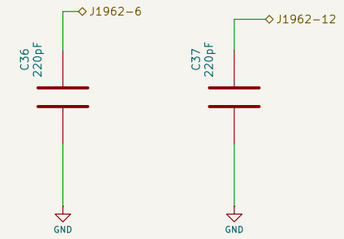
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

PCI / J1850



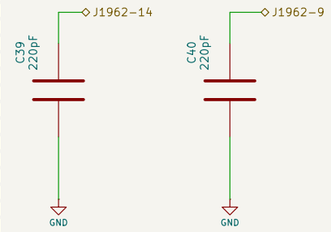
C25 — OBD pin 2

SCI TX6 / TX12



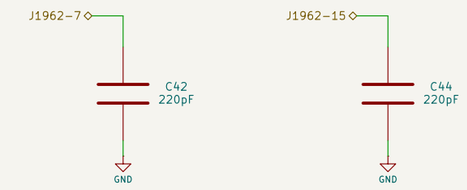
C36: pin 6 C37: pin 12

SCI TX14 / TX9



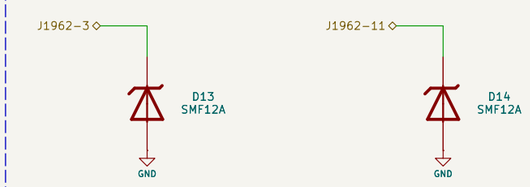
C39: pin 14 C40: pin 9

SCI RX7 / RX15



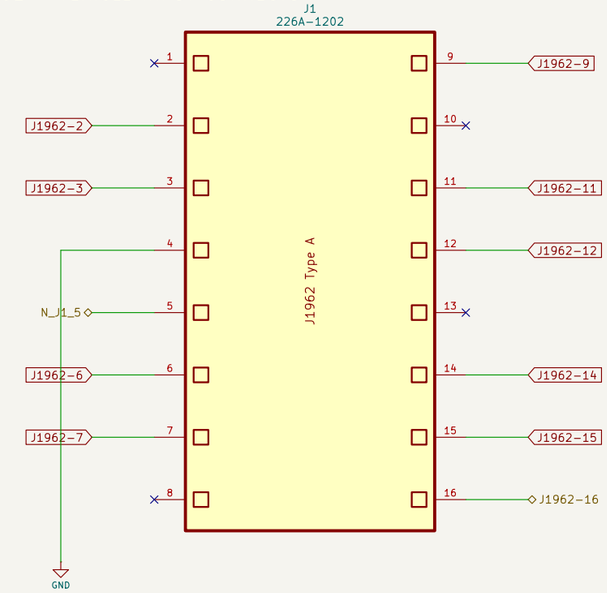
C42: pin 7 C44: pin 15

CCD bus protection



D13: pin 3 D14: pin 11

J1 — VEHICLE INPUT CONNECTOR



ChryslerScanner / atopile

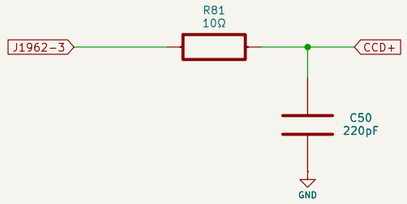
Sheet: /OBD connector + protection/
File: 06_obd-protection.kicad_sch

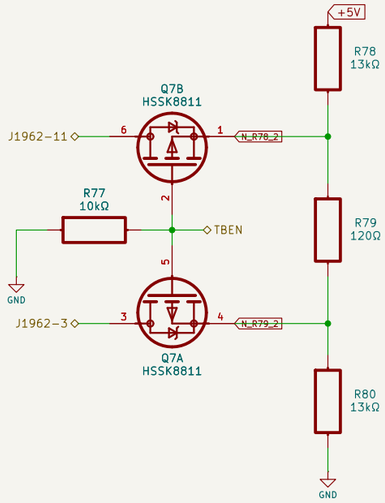
Title: OBD connector and line protection

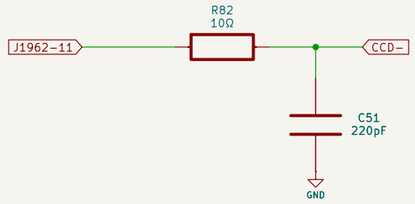
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 2/24

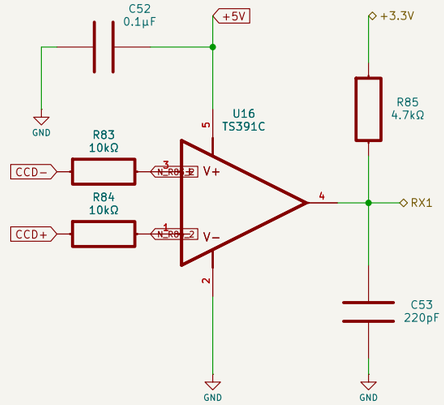
CCD interface

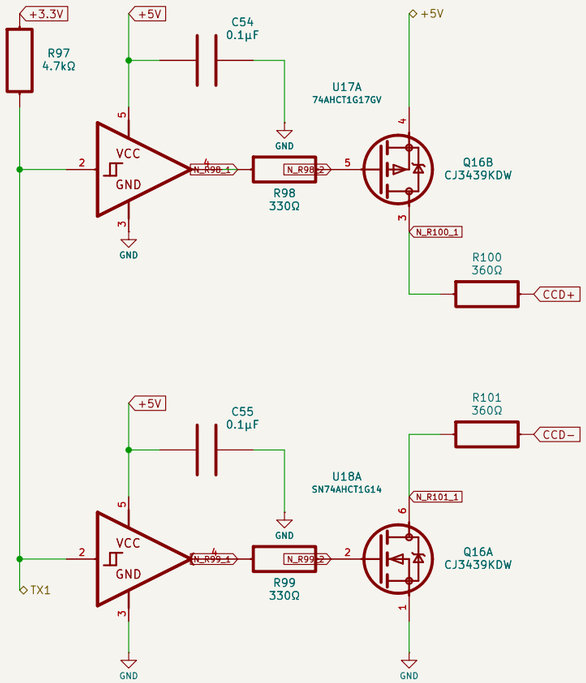
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

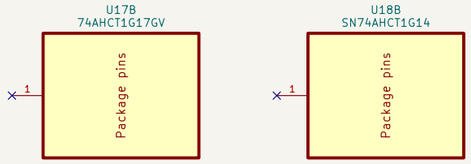












Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

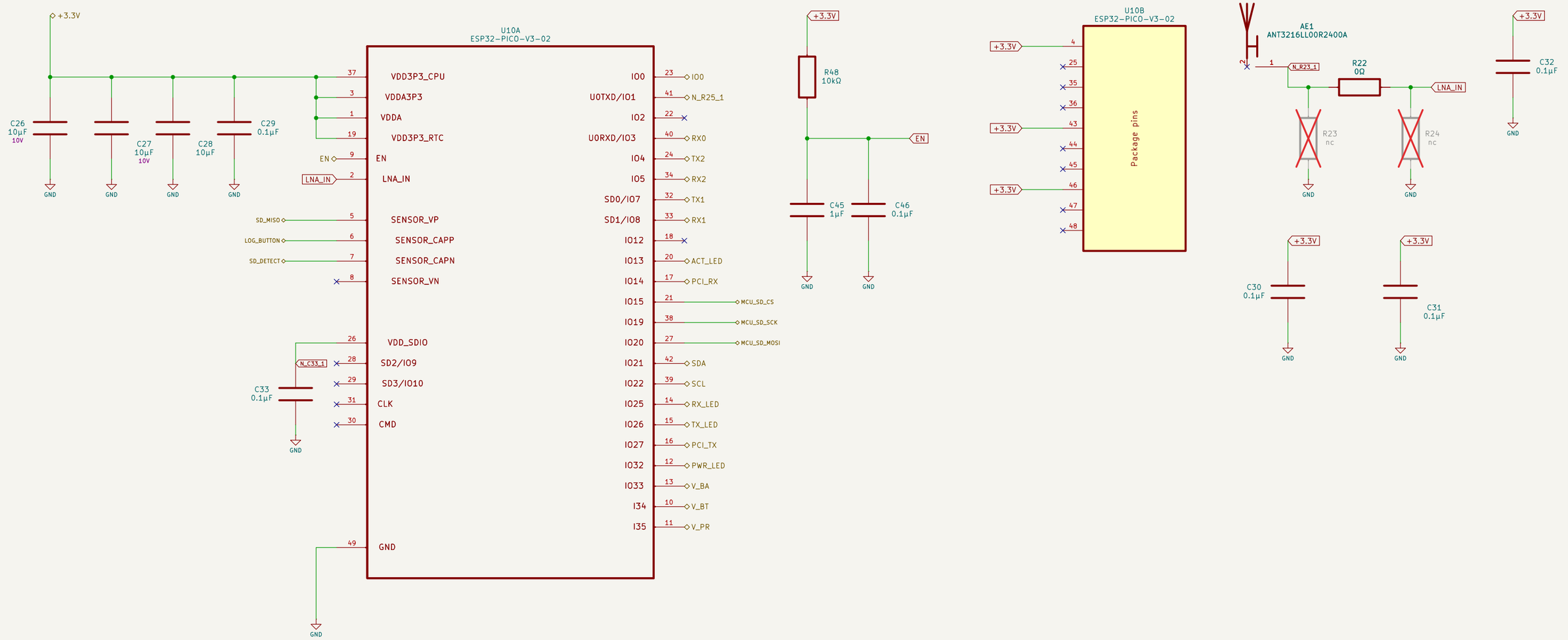
Sheet: /CCD interface/
File: 01_ccd.kicad_sch

Title: CCD interface

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 3/24

ESP32 + RF

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



SD revision: GPIO19/20 SPI, GPIO36 MISO, GPIO15 CS, GPIO37 button, GPIO38 detect. Firmware deferred.

Assembly option: R23 and R24 are DNP.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

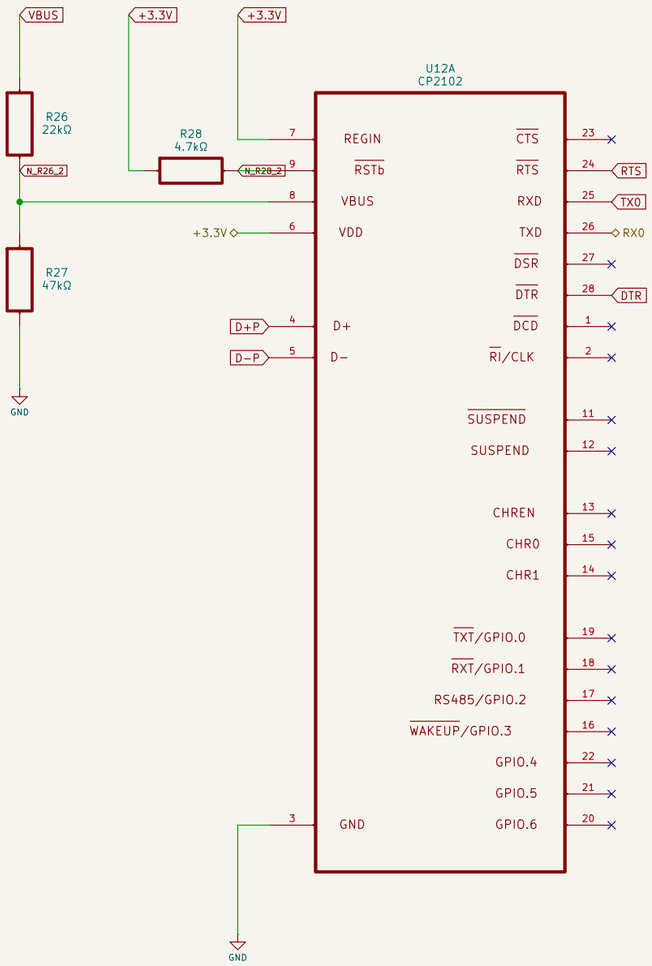
Sheet: /ESP32 + RF/
File: 16_mcu.kicad_sch

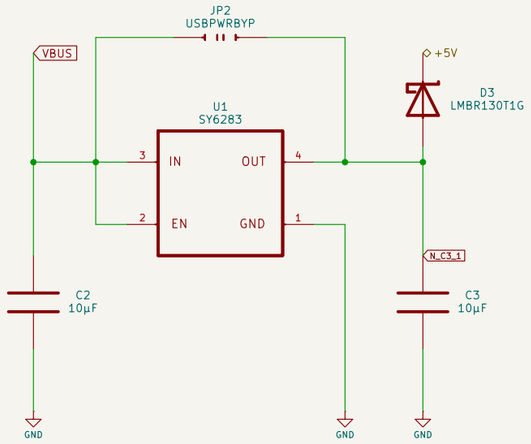
Title: ESP32 + RF

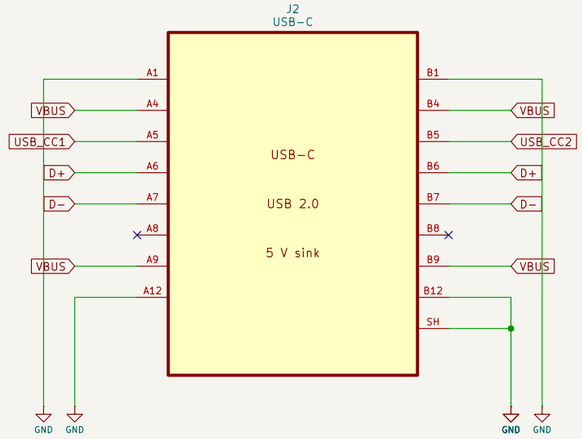
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 4/24

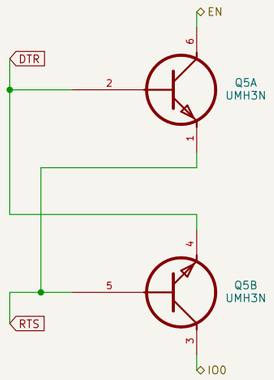
USB + reset / boot

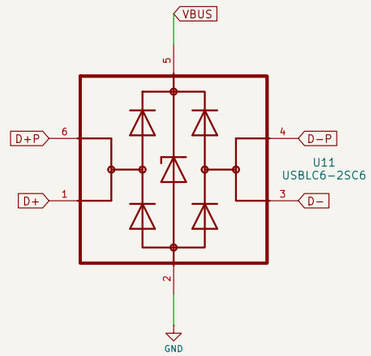
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

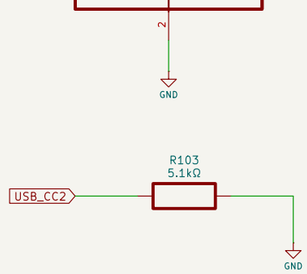


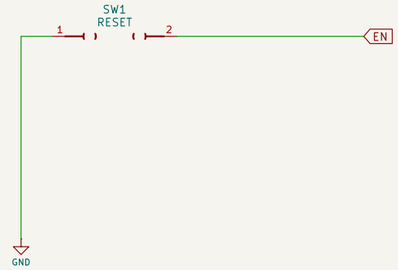


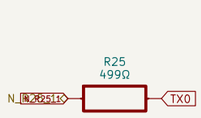


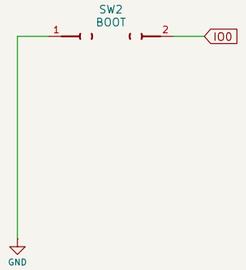


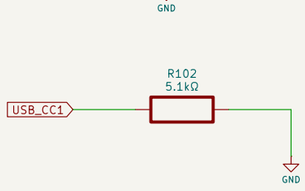


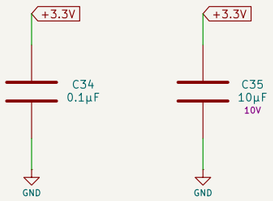


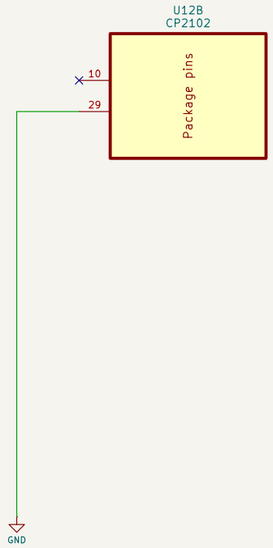












USB−C: USB 2.0 device / 5 V sink. R102 and R103: independent 5.1 kohm CC pull−downs.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

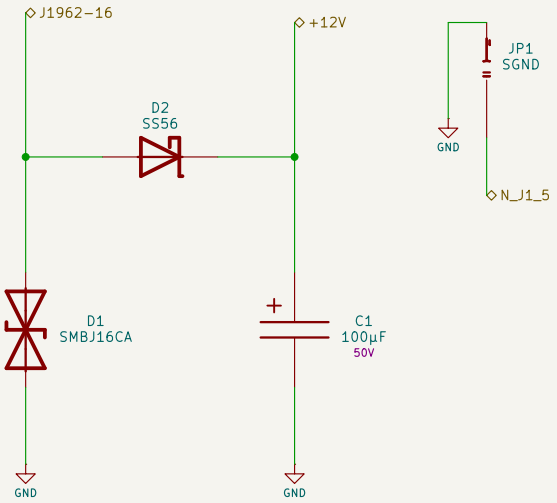
Sheet: /USB + reset / boot/
File: 19_usb.kicad_sch

Title: USB + reset / boot

Size: A3	Date: 2026−10−13	**Rev: V240−C/OBD**
KiCad E.D.A. 10.0.5		Id: 5/24

Vehicle power input

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

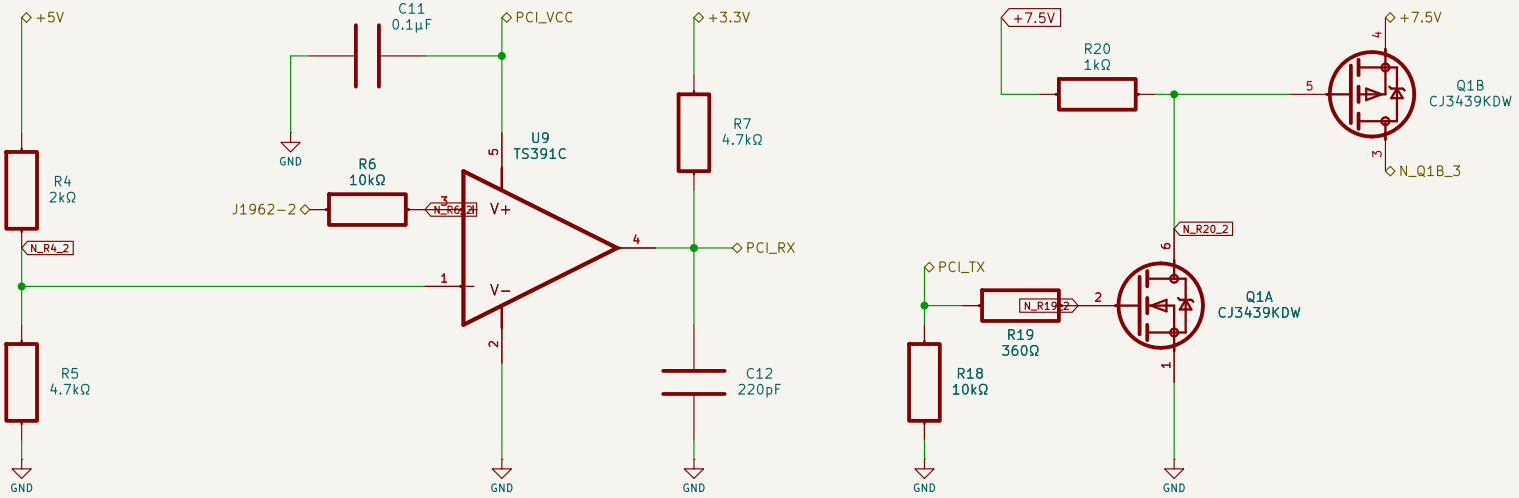
Sheet: /Vehicle power input/
File: 07_vehicle-input.kicad_sch

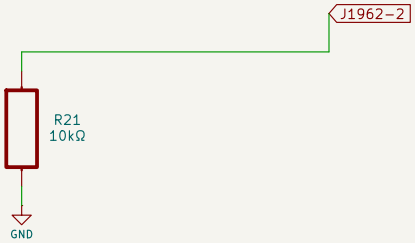
Title: Vehicle power input

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 6/24

PCI / J1850

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



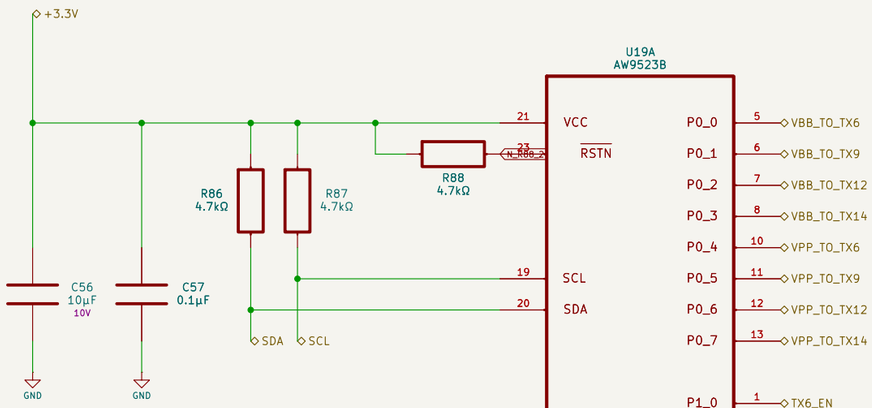


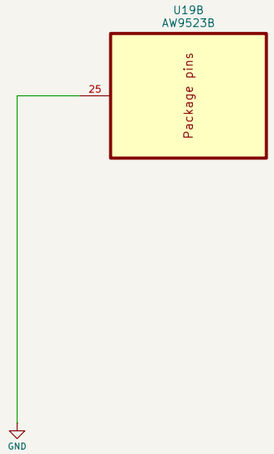
Atopile source / editable KiCad view / passive pins require electrical review.

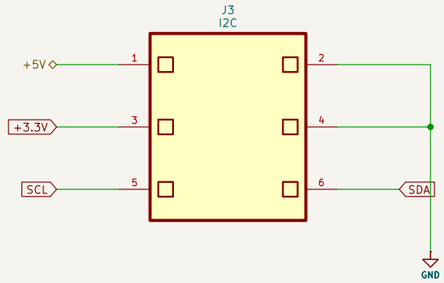
ChryslerScanner / atopile		
Sheet: /PCI / J1850/		
File: 02_pci.kicad_sch		
Title: PCI / J1850		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 7/24

I/O expander + I2C

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.







Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /I/O expander + I2C/
File: 18_control.kicad_sch

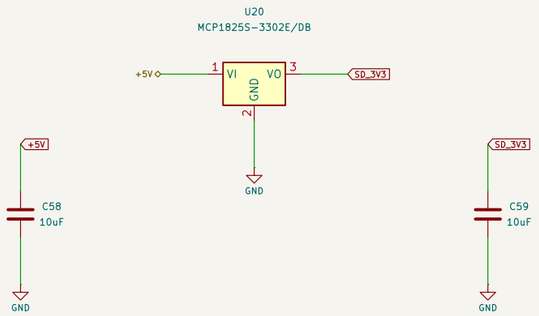
Title: I/O expander + I2C

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 8/24

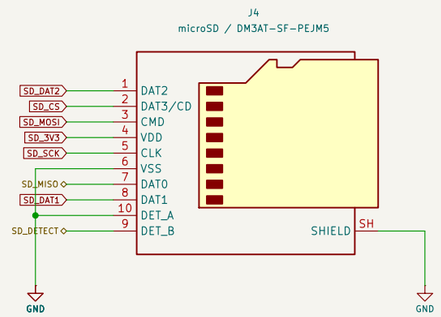
On-board microSD logger (SPI)

Existing ESP32-PICO-V3-02 | removable storage | native scanner functions retained | firmware deferred

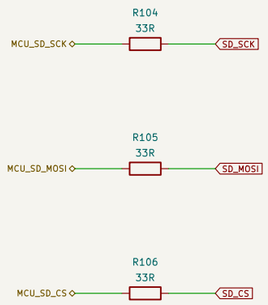
Dedicated 3.3 V SD supply from the existing +5 V rail



J4: Hirose DM3AT-SF-PEJM5 / SPI mode



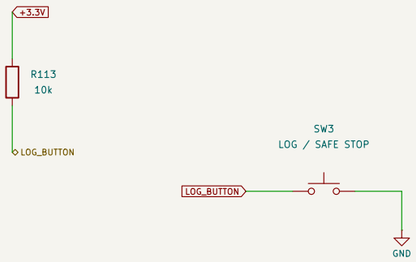
MCU-side series damping



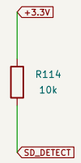
SD CMD/DAT pull-ups / MCU CS strap bias



Active-low record / safe-stop input



Card-detect input



GPIO15 CS is biased high at reset; GPIO12 and GPIO2 remain unused.

GPIO36/37/38 are input-only and have external pull-ups.

GPIO20 is package-specific; use ESP-IDF support for ESP32-PICO-V3.

SD rail transient current and total +5 V budget need hardware validation.

Bulk capacitance is not a safe-power-off hold-up supply.

Firmware deferred: this hardware revision does not yet record data.

ACT_LED / GPIO13 is reserved for the supervised health heartbeat.
No SD/activity writers, autonomous blink, PWM or DMA may drive this pin.

Existing sheets use passive visualization pins; native ERC is not independent certification of the scanner circuit.

ChryslerScanner / local hardware revision

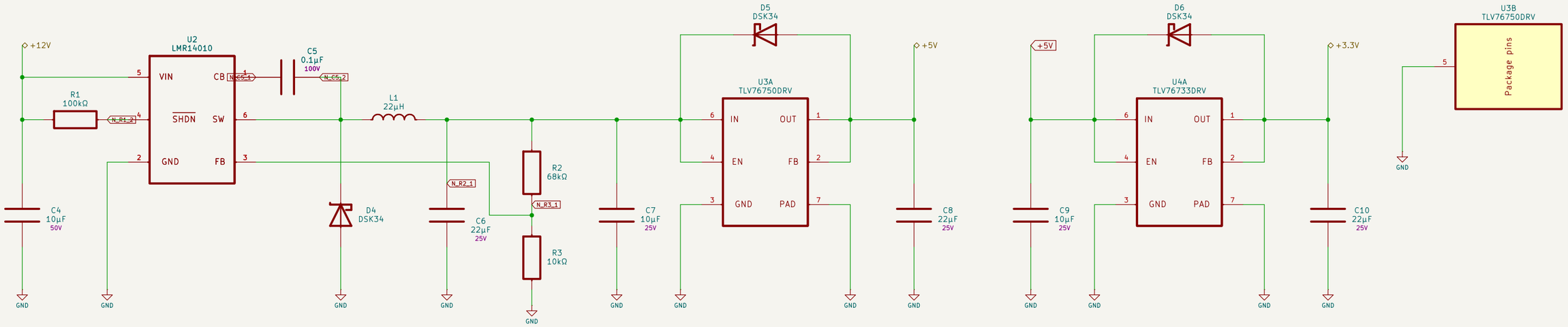
Sheet: /Integrated SD logger/
File: 22_sd-logger.kicad_sch

Title: Integrated microSD logging hardware

Size: A3	Date: 2026-10-20	**Rev: V240-SD-A**
KiCad E.D.A. 10.0.5		Id: 9/24

5 V / 3.3 V supplies

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



All TLV767 ground terminals and exposed pads connect to GND in this checked revision.

Native schematic and PCB ground connectivity are checked in the final validation reports.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

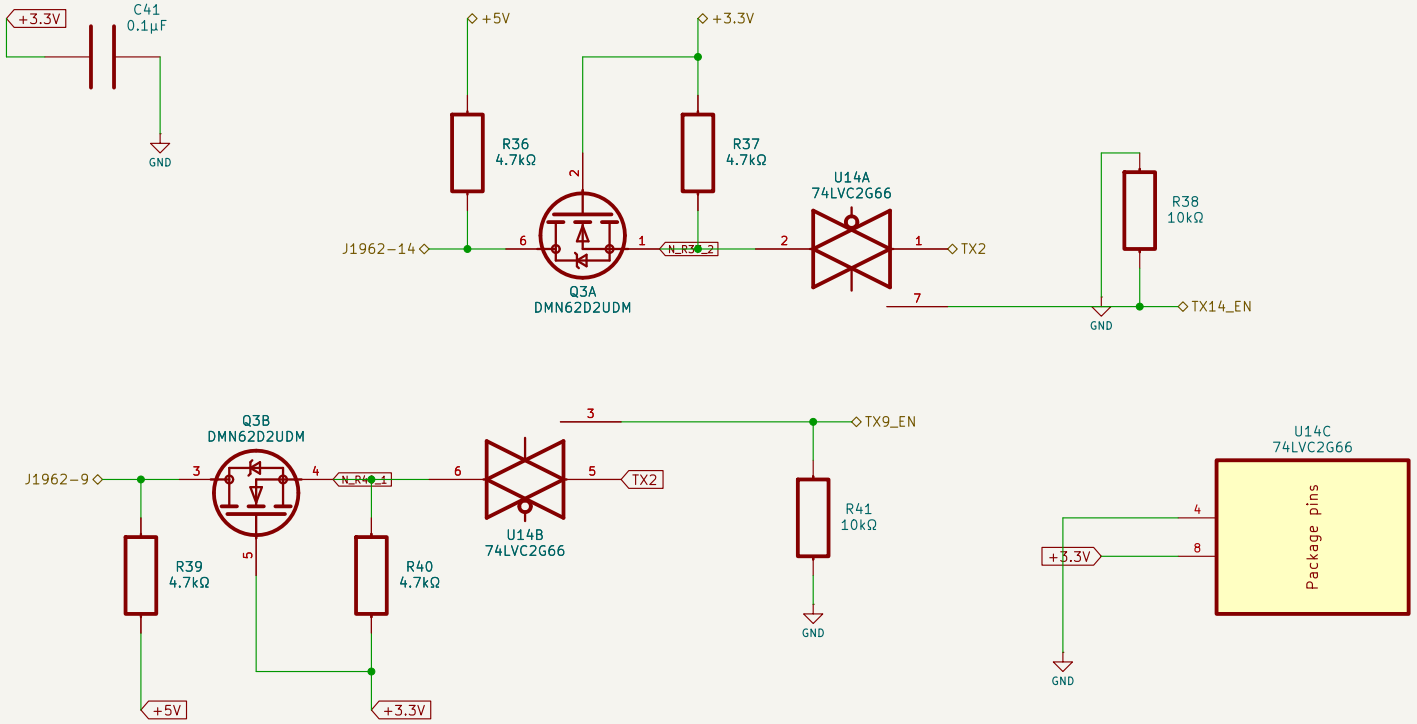
Sheet: /5 V / 3.3 V supplies/
File: 08_power-5v.kicad_sch

Title: 5 V / 3.3 V supplies

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 10/24

SCI TX14 / TX9

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

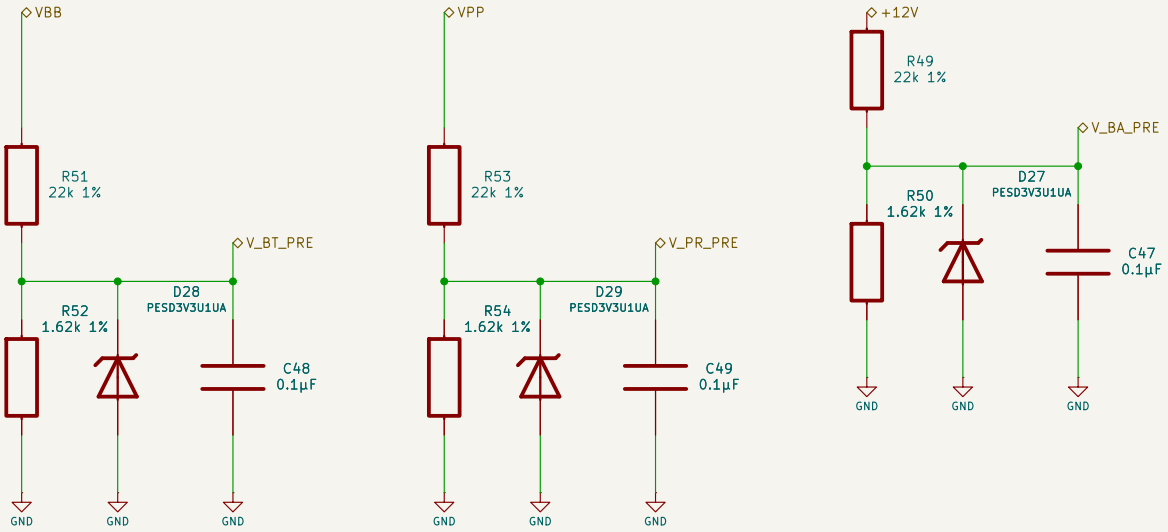


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI TX14 / TX9/		
File: 03_sci-14-9.kicad_sch		
Title: SCI TX14 / TX9		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 11/24

Voltage monitoring

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Rev C: 22k / (1.62k || 100k), nominal ADC/input gain = 0.06757.
Calibrate switch leakage/resistance, tolerances and ADC response.
V_BA senses protected +12V after D2; connector voltage needs diode-drop correction.
Original 8.66k divider conversion constants do not apply.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

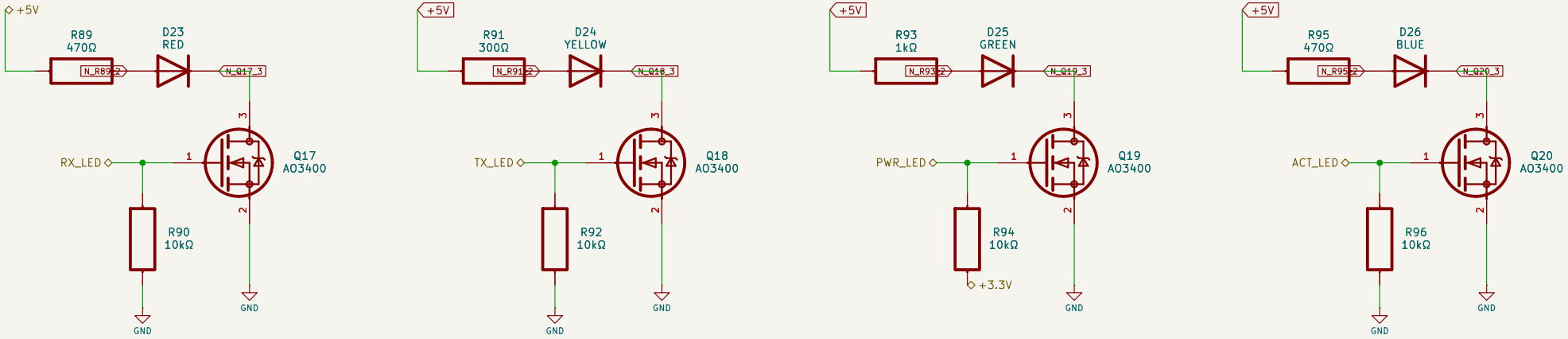
Sheet: /Voltage monitoring/
File: 17_adc.kicad_sch

Title: Voltage monitoring

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 12/24

Status LED drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

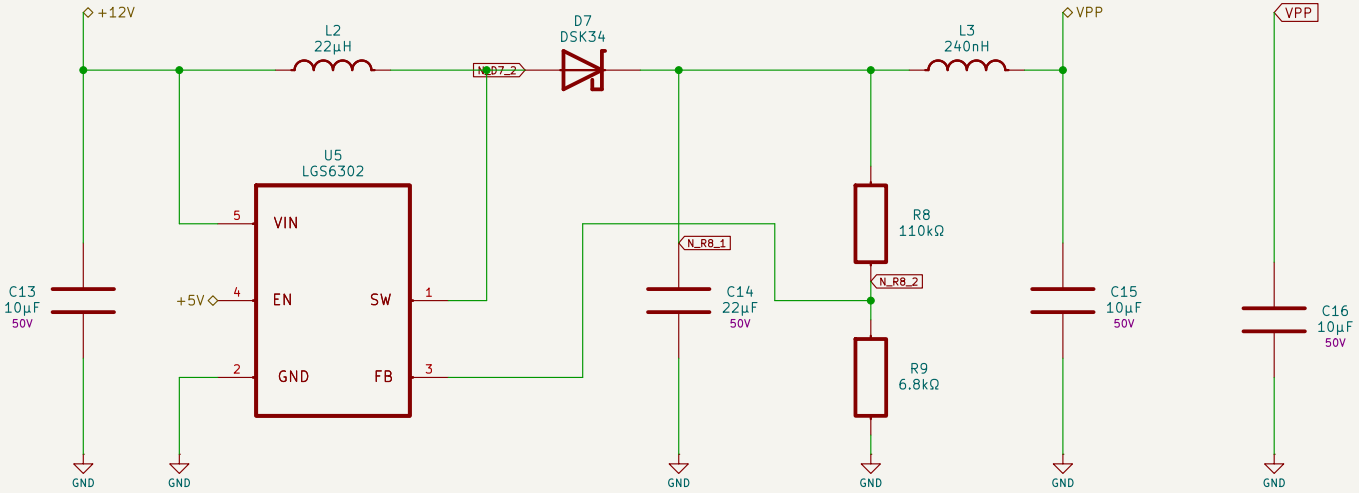
Sheet: /Status LED drivers/
File: 20_leds.kicad_sch

Title: Status LED drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 13/24

VPP boost supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

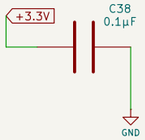
Sheet: /VPP boost supply/
File: 09_power-vpp.kicad_sch

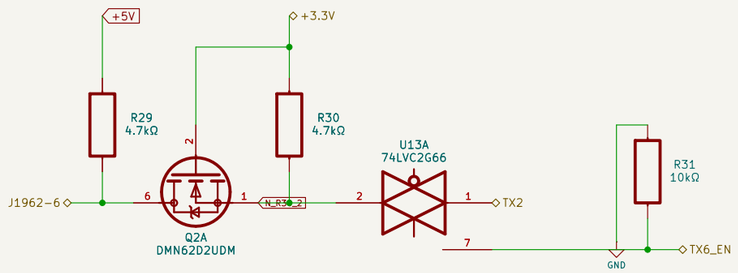
Title: VPP boost supply

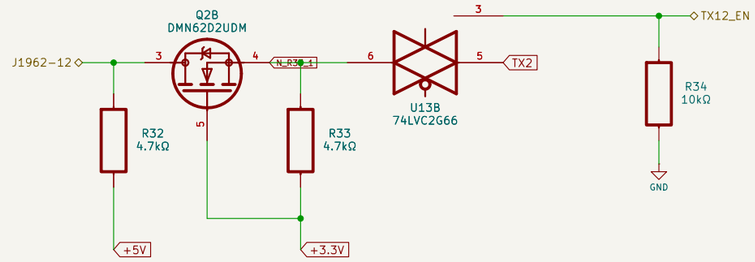
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 14/24

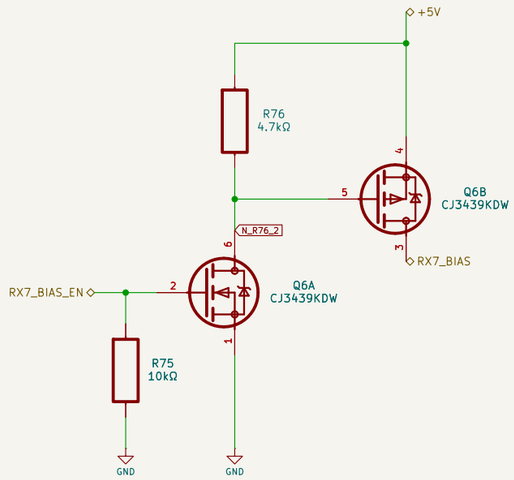
SCI TX6 / TX12 + RX bias

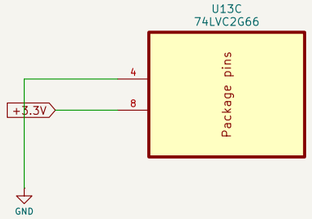
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.











Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

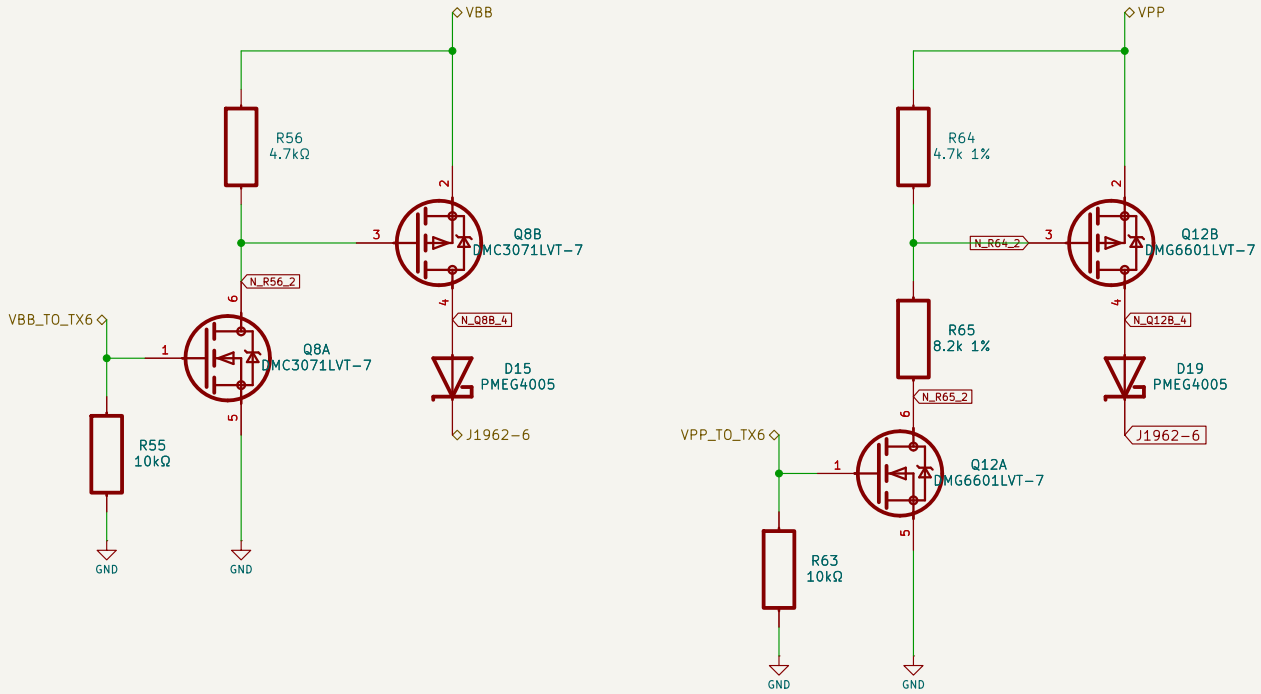
Sheet: /SCI TX6 / TX12 + RX bias/
File: 04_sci-6-12.kicad_sch

Title: SCI TX6 / TX12 + RX bias

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 15/24

SCI TX6 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

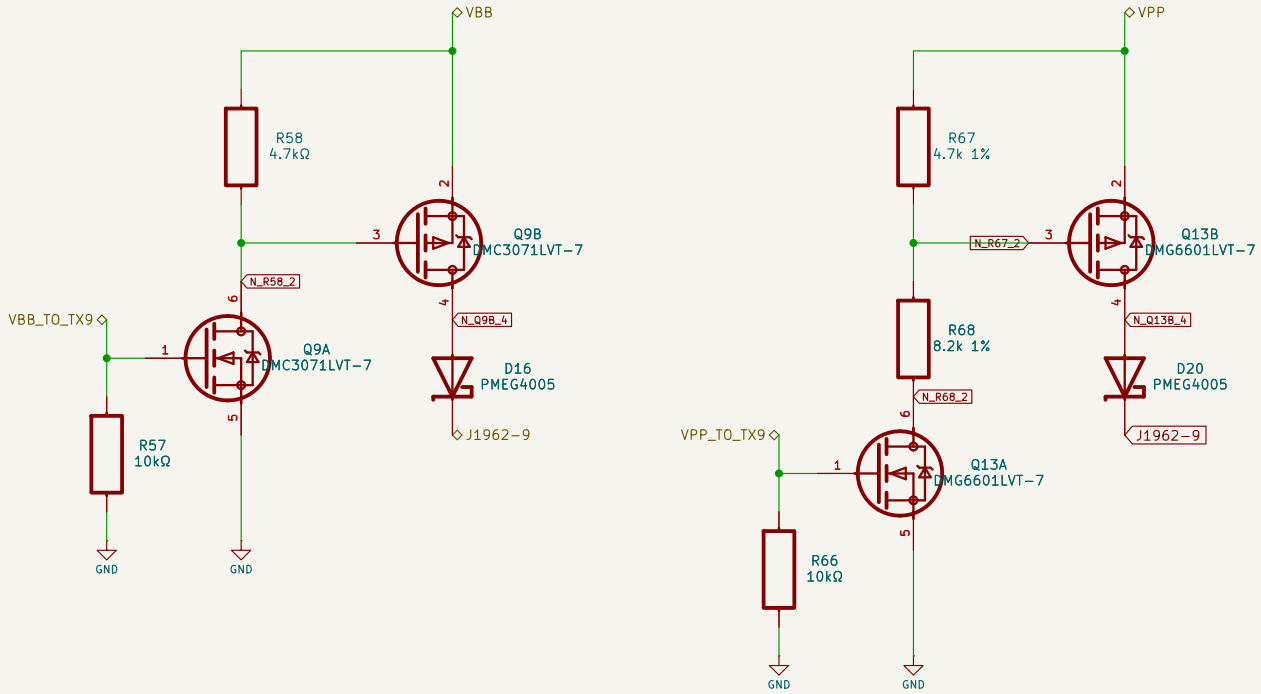
Sheet: /SCI TX6 VBB / VPP drivers/
File: 12_program-6.kicad_sch

Title: SCI TX6 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 16/24

SCI TX9 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

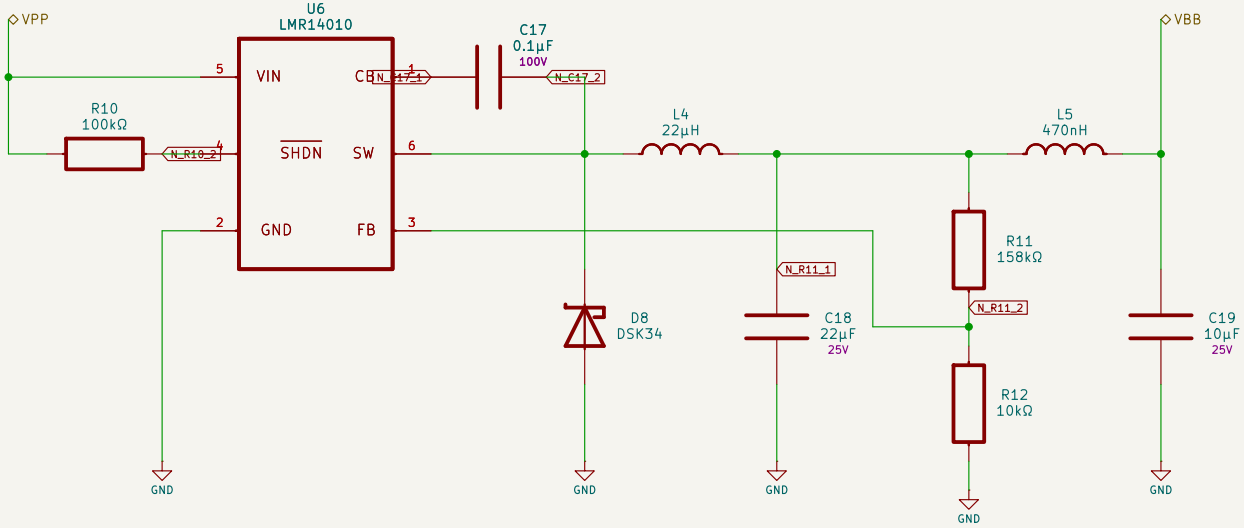
Sheet: /SCI TX9 VBB / VPP drivers/
File: 13_program-9.kicad_sch

Title: SCI TX9 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 17/24

VBB supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

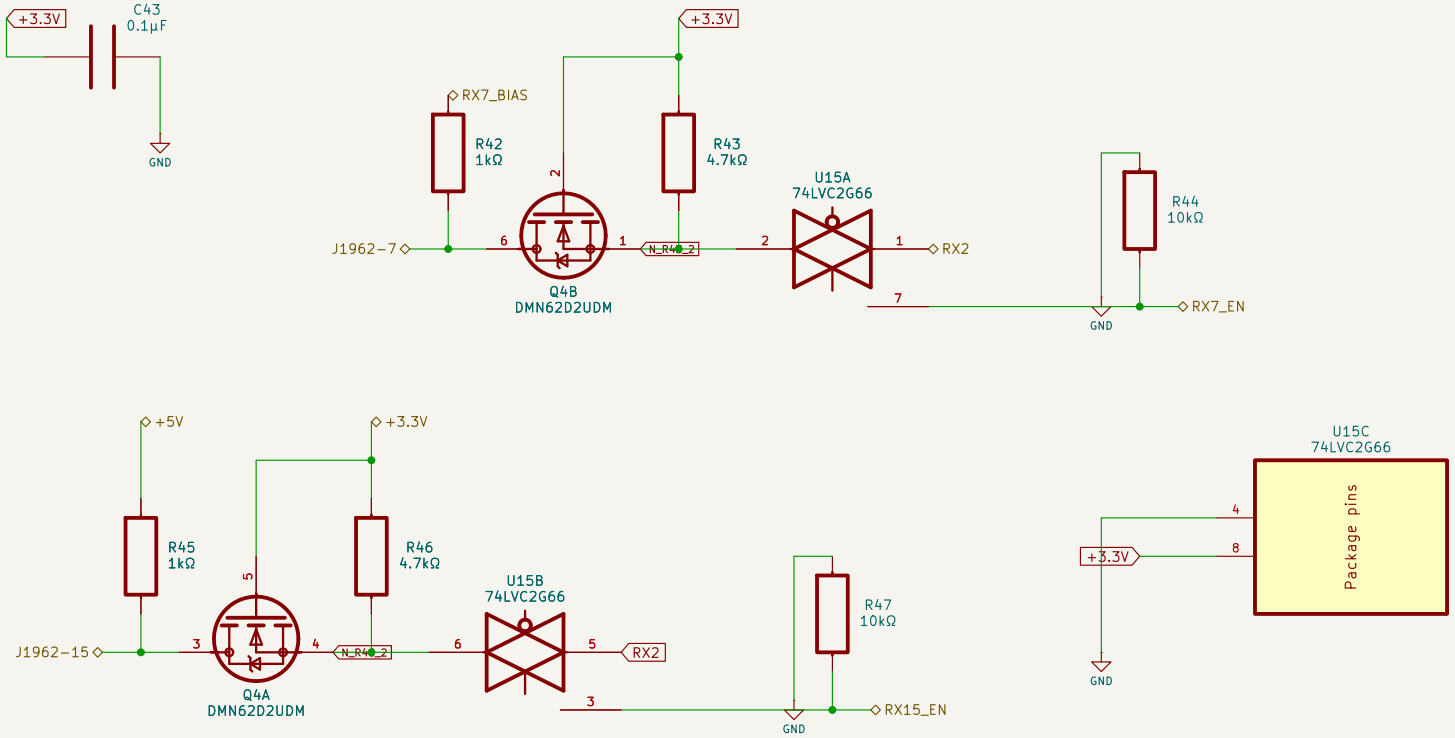


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /VBB supply/ File: 10_power-vbb.kicad_sch		
Title: VBB supply		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 18/24

SCI RX7 / RX15

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

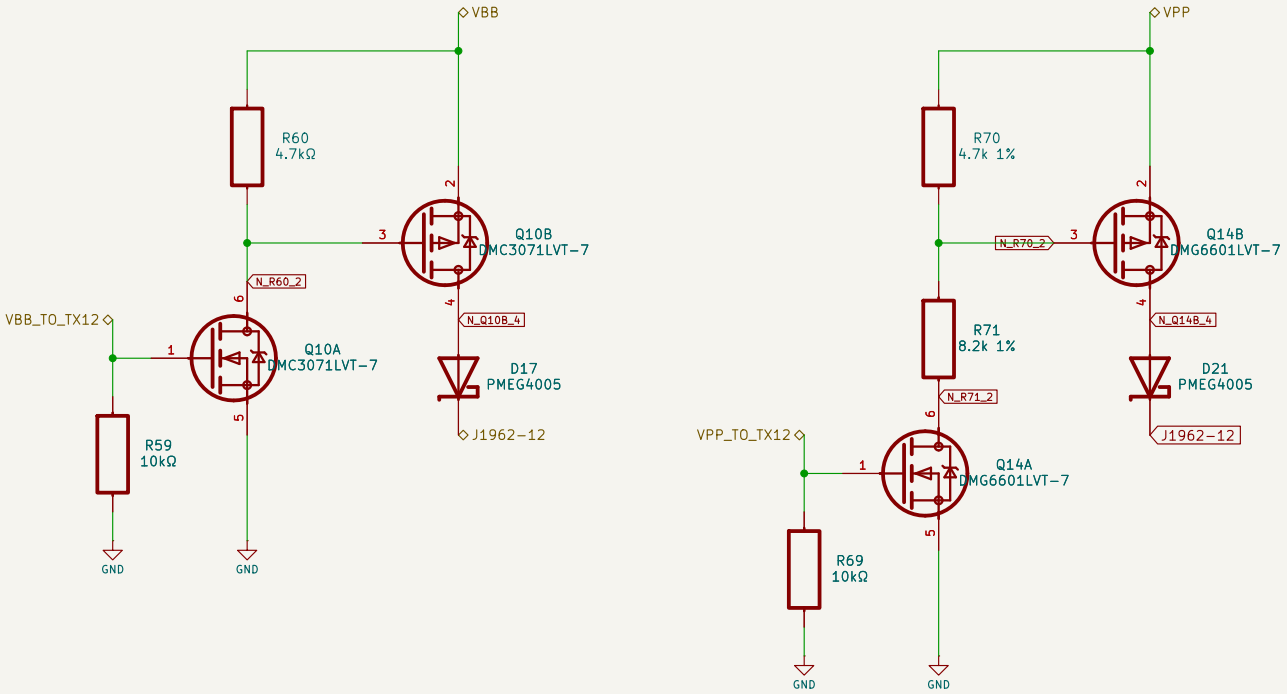


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI RX7 / RX15/		
File: 05_sci-rx.kicad_sch		
Title: SCI RX7 / RX15		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 19/24

SCI TX12 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

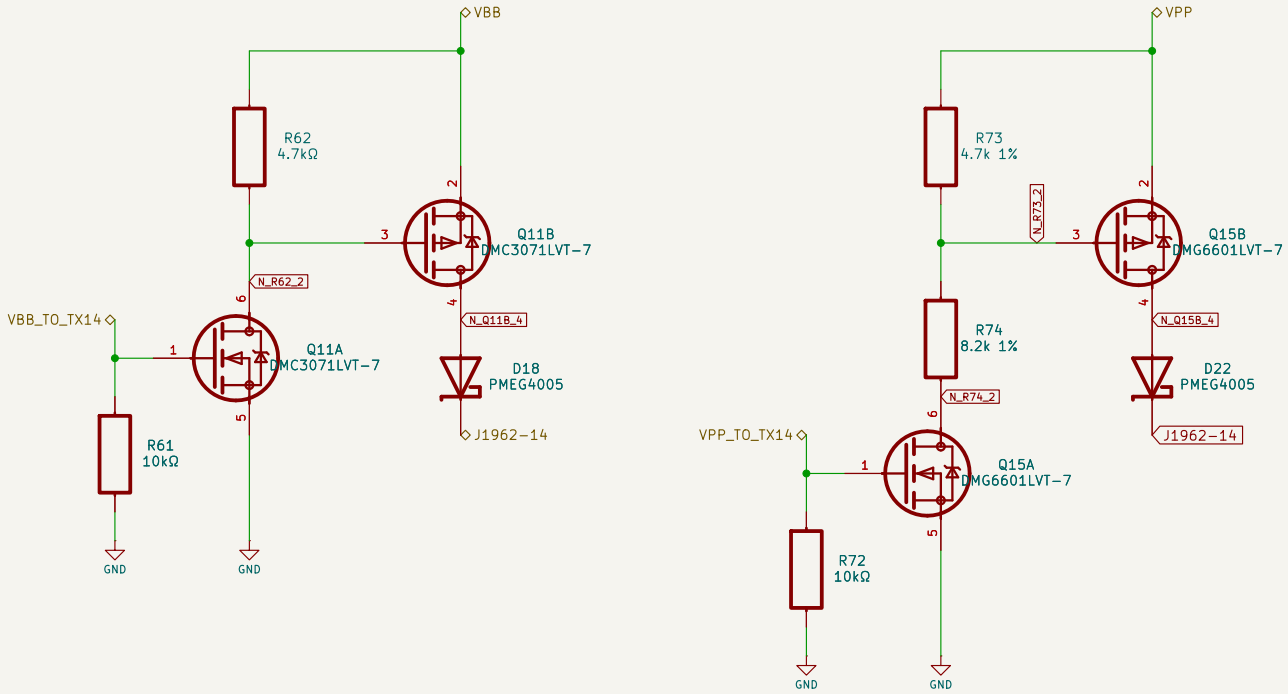
Sheet: /SCI TX12 VBB / VPP drivers/
File: 14_program-12.kicad_sch

Title: SCI TX12 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 20/24

SCI TX14 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

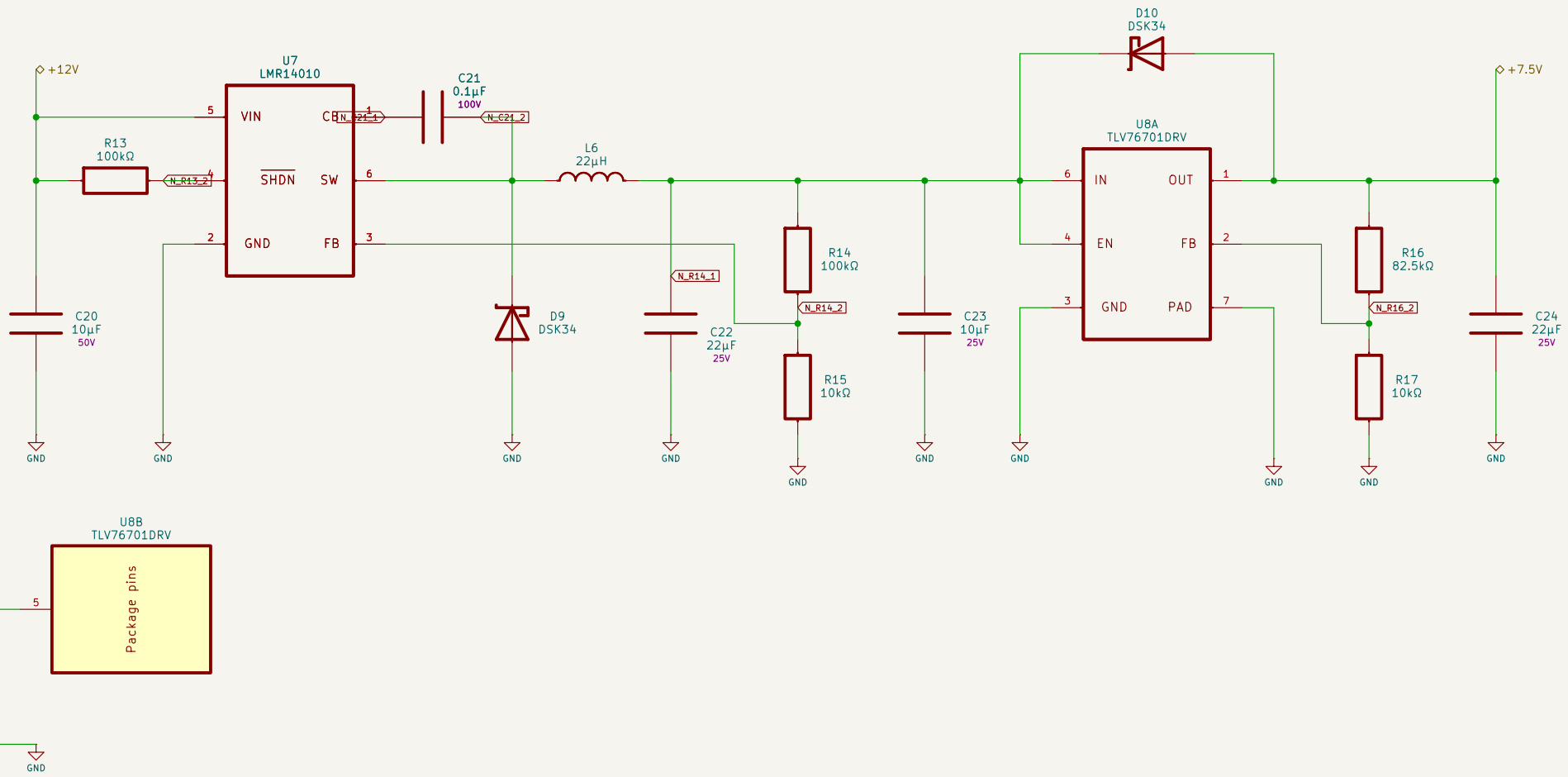


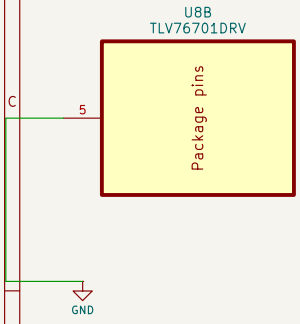
Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI TX14 VBB / VPP drivers/		
File: 15_program-14.kicad_sch		
Title: SCI TX14 VBB / VPP drivers		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 21/24

7.5 V supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.





All TLV767 ground terminals and exposed pads connect to GND in this checked revision.

Native schematic and PCB ground connectivity are checked in the final validation reports.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /7.5 V supply/
File: 11_power-7v5.kicad_sch

Title: 7.5 V supply

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 22/24

ADC isolation / supply supervisor / heartbeat-only I/O expander permit

+3.3V +12V PCI_VCC V_BA_PRE V_BT_PRE V_PR_PRE V_BA V_BT V_PR ACT_LED EN N_R88_2

U21
TMUX1511PWR
LOGIC_PG 1
V_BA_PRE 2
V_BA 3
LOGIC_PG 4
V_BT_PRE 5
V_BT 6
7
8 V_PR
9 V_PR_PRE
10 LOGIC_PG
11
12
13
14 +3.3V
GND GND GND GND

U22
LTC6993HS6-2#TRMPBF
ACT_LED 1
2
WD_SET 3
4 WD_DIV
5 +3.3V
6 HEARTBEAT_VALID
GND

U23
SN74LVC1G11DBVR
EN 1
2
LOGIC_PG 3
4 N_R88_2
5 +3.3V
6 HEARTBEAT_VALID
GND

U24
TLV803EA30DBZR
1
LOGIC_PG 2
3 +3.3V
GND

R115
100k 0.1% 25ppm
WD_SET 1 2 2
GND

R116
191k 1%
+3.3V 1 2 2 WD_DIV

R117
100k 1%
WD_DIV 1 2 2
GND

R118
10k 1%
ACT_LED 1 2 2
GND

R119
10k 1%
+3.3V 1 2 2 LOGIC_PG

R120
100k 1%
V_BA 1 2 2
GND

R121
100k 1%
V_BT 1 2 2
GND

R122
100k 1%
V_PR 1 2 2
GND

R123
1k 1W 1%
+12V 1 2 2 PCI_VCC

D30
BZT52H-B15
PCI_VCC 1 2 2
GND

C62
100nF 16V X7R
+3.3V 1 2 2
GND

C63
100nF 16V X7R
+3.3V 1 2 2
GND

C64
100nF 16V X7R
+3.3V 1 2 2
GND

C65
100nF 16V X7R
+3.3V 1 2 2
GND

+3.3V

GPIO13 / ACT_LED is dedicated to software health pulses every 10 ms; no activity-only source or PWM.

R49 senses protected input after D2; all ADC conversion factors change. See rev-C safety contract.

Sheet: /Hardware safety + ADC isolation/
File: 23_hardware-safety.kicad_sch

Title: Hardware safety / ADC isolation

Size: A3	Date: 2026-10-21	**Rev: V240-SD-C**
KiCad E.D.A. 10.0.5		Id: 23/24

DEBUG ACCESS — TP1-TP52

5015 SMT hook loops; TP45-TP48 are bare SPI probe pads. Numbered silkscreen references map to nets here.

TP1	J1962-16	TP14	J1962-6
TP2	+12V	TP15	J1962-12
TP3	VBUS	TP16	J1962-14
TP4	+5V	TP17	J1962-9
TP5	+3.3V	TP18	J1962-7
TP6	SD_3V3	TP19	J1962-15
TP7	+7.5V	TP20	TX0
TP8	VBB	TP21	RX0
TP9	VPP	TP22	EN
TP10	PCI_VCC	TP23	IO0
TP11	J1962-2	TP24	SCL
TP12	J1962-3	TP25	SDA
TP13	J1962-11	TP26	PCI_RX
TP27	PCI_TX	TP40	V_PR_PRE
TP28	RX1	TP41	V_PR
TP29	TX1	TP42	SD_DETECT
TP30	RX2	TP43	CCD+
TP31	TX2	TP44	CCD−
TP32	ACT_LED	TP45	SD_CS
TP33	HEARTBEAT_VALID	TP46	SD_SCK
TP34	LOGIC_PG	TP47	SD_MOSI
TP35	N_R88_2	TP48	SD_MISO
TP36	V_BA_PRE	TP49	GND
TP37	V_BA	TP50	GND
TP38	V_BT_PRE	TP51	GND
TP39	V_BT	TP52	GND

Sheet: /Debug access / test points/	
File: 24_debug-testpoints.kicad_sch	
Title: Debug access / hook test points	
Size: A3	Date: 2026-10-22
KiCad E.D.A. 10.0.5	**Rev: V240-SD-C**
	Id: 24/24